Whitepaper

TradingView Pre-Market In-Play Scanner

What the Pine Script does, what it needs, when to run it, and how to interpret the results.

Script type	**TradingView Pine Script indicator designed to score pre-market candidates**
Primary use	Build a focused watchlist of in-play names and isolate likely avoids
Best run window	U.S. pre-market, using a 5m or 15m chart / screener interval

One-sentence summary

This script does not magically scan the whole market by itself. It scores the symbols you feed into it, tags which ones look genuinely in-play, ranks them by tradability, and pushes thin or messy names into a Do Not Trade bucket.

Version described: the Pine Script provided in this chat thread, including its current defaults and hard-coded scoring rules.

Prepared for the user • 2026-03-08

Executive Summary

The Pine Script is a pre-market scoring and triage tool. It is meant to help you take a pool of candidate stocks, identify which ones are active enough to matter, rank the better trading candidates, and quarantine the names that are likely to be too thin, too low-priced, or too structurally messy to trade safely.

Conceptually, the script follows the workflow in your prompt. It looks at pre-market activity, applies an 'in-play' test using a relative-volume proxy, calculates a tradability score, and separately calculates an avoid score. The best names are the ones that pass the in-play test, score well on tradability, and do not trip the avoid logic.

A very important nuance is that the current script does not use a native vendor-supplied RVOL field. Instead, it implements the fallback branch from your prompt: a stock passes only if pre-market volume is at least 10% of average daily volume, or if pre-market dollar volume is at least $1,000,000.

Another important nuance is that the script does not generate exactly eight tradable names and five avoid names by itself. It produces columns and scores. You then sort and filter the scan results to extract the top eight tradable names and the five strongest avoids.

Bottom line
Use this script after you have already assembled a list of pre-market movers. It is best understood as a ranking and filtering layer, not a complete market-wide discovery engine.

Purpose and Scope

The purpose of the script is to compress a noisy pre-market universe into a more practical trading list. In pre-market trading, many stocks may be moving, but only a subset are truly liquid enough, clean enough, and active enough to be worth attention. This script tries to separate those groups.

The script is built around three practical questions:

- Is this stock active enough to count as in-play?
- If it is active, how tradable does it look based on liquidity, directional clarity, and room around key levels?
- Even if it is moving, is it thin or messy enough that it belongs in a Do Not Trade section?

The scope is specifically U.S. pre-market behavior. The default session input is 04:00 to 09:29 New York time, Monday through Friday. The script is also constrained to intraday timeframes of 30 minutes or less.

```
tfOK = timeframe.isintraday and timeframe.in_seconds() <= 1800
```

How the Script Works

1. Collect pre-market statistics: The script builds an extended-hours ticker, then accumulates pre-market volume, high, low, opening price, and the latest pre-market price during the configured pre-market session.

2. Pull daily context: It fetches the prior day high, low, and close, plus average daily volume and ATR from daily data. These are used as anchors for comparison and scoring.

3. Apply the in-play proxy: A symbol is considered in-play only if it clears the proxy threshold: pre-market volume as a percent of ADV, or pre-market dollar volume.

4. Score tradability: The script converts liquidity, gap size, directional cleanliness, and distance from key daily levels into a 0-100 tradability score.

5. Score avoid risk: It separately adds penalties for low pre-market dollar volume, thin average daily dollar volume, low price, messy pre-market structure, and failure of the in-play proxy.

Internally, the pre-market accumulation happens inside a helper function named f_pmStats(). That function identifies whether the current bar is inside the configured pre-market session and, when it is, it continually updates cumulative pre-market volume, pre-market high, pre-market low, and pre-market open.

The function also computes a directional-cleanliness measure called trend efficiency. In plain language, trend efficiency asks whether the price moved through the pre-market range in a relatively clean directional way, or whether it chopped around inside that range.

```
trendEfficiency = abs(close - pmOpen) / (pmHigh - pmLow)
```

A higher trend-efficiency reading means the move was cleaner. A lower reading means the pre-market action was more back-and-forth and less directional.

User Inputs and Defaults

All visible settings in the script are optional because the script ships with defaults. You can run it immediately without changing anything. That said, changing these settings can materially change the candidate list you get.

Input	Default	What it controls	How it changes outcomes
pmSession	0400-0929:23456	Defines the pre-market session window the script will accumulate.	Widening the window increases counted pre-market activity; narrowing it makes the script stricter and later-session focused.
pmTZ	America/New_York	Timezone used when interpreting the pre-market session string.	Changing this incorrectly can shift the session and corrupt the pre-market calculations.

advLen	20	Lookback used for average daily volume (ADV).	A shorter lookback makes ADV adapt faster; a longer lookback smooths unusual recent volume.
minPmPctAdv	10.0	Minimum pre-market volume as a percent of ADV required to pass the proxy.	Raising it reduces candidates and increases strictness; lowering it allows weaker names through.
minPmDollarVol	$1,000,000	Alternative way to pass the proxy using pre-market dollar volume.	Raising it favors more liquid names; lowering it includes thinner names.
avoidPmDollarVol	$300,000	Minimum pre-market dollar volume before a low-liquidity risk flag appears.	Higher values push more names into the avoid bucket.
avoidAdvDollarVol	$15,000,000	Minimum average daily dollar volume before a thin-liquidity risk flag appears.	Higher values make the tool more biased toward liquid large-cap or mid-cap names.
avoidPrice	$2.00	Minimum price before a low-price risk flag appears.	Lowering it admits more cheap stocks; raising it excludes more low-priced names.

Required vs. optional

From a coding standpoint, none of the visible inputs are mandatory because defaults are provided. In practice, the only thing you must provide operationally is a universe of candidate symbols to scan. The script cannot evaluate stocks you do not feed into it.

Derived Metrics and Internal Variables

After collecting the raw data, the script derives several intermediate metrics. These are not user inputs, but they directly drive the final scores.

Variable	Meaning	Why it matters
pmDollarVol	Pre-market volume multiplied by the latest available price.	Used for the pass/fail proxy and for liquidity scoring.
advDollarVol	Average daily volume multiplied by previous close.	Used as a baseline measure of normal liquidity.
pmVolPctAdv	Pre-market volume divided by ADV, expressed as a percent.	One branch of the in-play proxy.
gapPct	Percent gap between the latest pre-	Feeds the catalyst proxy.

	market price and the previous close.	
pmRangePct	Pre-market range as a percent of pre-market open.	Helps detect overextension and messy structure.
nearestDailyLevelRoomPct	Distance from current price to the nearest of prior high, low, or close, expressed as a percent of price.	Approximates how much room price has before colliding with a nearby daily reference level.
atrPct	Daily ATR divided by previous close, expressed as a percent.	Used to penalize unusually stretched pre-market ranges.

The in-play gate is binary. A name either passes or it does not. In the current code, the gate is:

```
proxyPass = (pmVolPctAdv >= minPmPctAdv) OR (pmDollarVol >= minPmDollarVol)
```

This is the script's core answer to the question 'Is this actually in play?' If the answer is no, the stock can still appear in the scan, but it will receive a much lower tradability score and a risk flag for failing the proxy.

Tradability Score Model

Tradability is a weighted blend of three ideas: liquidity, catalyst-style price action, and structural cleanliness around nearby levels. The final score is normalized to a 0-100 style scale.

Component	Weight in tradability	Interpretation
Liquidity score	50% of tradability	Built from pre-market dollar volume, average daily dollar volume, and price. Higher liquidity generally improves fills and reduces spread risk.
Catalyst proxy	25% of tradability	Built from absolute gap size and trend efficiency. The script uses price/volume behavior as a stand-in for catalyst clarity.
Clean levels	25% of tradability	Rewards room from nearby daily levels and penalizes names that are too stretched or too disorderly in pre-market action.

The actual tradability formula is:

```
tradabilityRaw = 0.50 * liquidityScore + 0.25 * catalystProxy + 0.25 * cleanLevels
```

If the in-play proxy fails, the script applies a major haircut:

```
tradability = proxyPass ? tradabilityRaw : tradabilityRaw * 0.25
```

That haircut is important. It means a stock might still look interesting on gap size alone, but if it does not clear the volume/liquidity proxy, the script intentionally downgrades it hard.

Avoid Score and 'Do Not Trade' Logic

Separately from tradability, the script adds an avoid score. This is the mechanism that populates the Do Not Trade section.

Risk flag	Condition	Penalty
riskLowPmDollar	Pre-market dollar volume below avoidPmDollarVol	+35
riskThinAdvDollar	Average daily dollar volume below avoidAdvDollarVol	+25
riskLowPrice	Price below avoidPrice	+20
riskMessyPM	Pre-market range above 8% while trend efficiency is below 0.35	+15
riskFailsProxy	The symbol did not pass the in-play proxy	+15

The script then classifies the name as an avoid when the summed penalty reaches 35 or more.

```
avoid = avoidScore >= 35
```

This design has a very practical consequence: a name with extremely low pre-market dollar volume can become an avoid immediately, even if nothing else is wrong, because the low pre-market liquidity penalty alone is 35.

Hard-Coded Assumptions That Also Affect Results

Not every important threshold is exposed as a user input. Several rules are hard-coded inside the script. These matter because changing them would require editing the code itself.

Hard-coded rule	Current value	Practical effect
Timeframe filter	Only intraday timeframes up to 30 minutes are valid.	If you run it on daily or slower charts, the script is effectively disabled.
Liquidity scaling bands	PM dollar score uses about $300k to $50M; ADV dollar score uses about $15M to $500M.	These bands control how quickly liquidity scores saturate.
Price score band	$2 to $20	Names above $20 do not keep gaining price-score credit from this subcomponent.
Gap score band	2% to 25%	Moves smaller than roughly 2% score weakly; huge gaps saturate.
Room score band	0.20% to 2.00%	Very small room to a daily level scores poorly.
Overextension penalties	10-point penalty above 12% PM	Very extended names are explicitly

	range; 20-point penalty above 20% PM range.	downgraded even when active.
ATR stretch penalty	10 points when PM range exceeds roughly 1.8x ATR percent.	Unusually stretched pre-market action is treated as less clean.
Tradable alert cutoff	60	The alertcondition considers a stock a tradable candidate at tradability >= 60, provided it passes the proxy and is not an avoid.

When to Run the Script

The ideal time to run the script is during the U.S. pre-market session, once enough volume has developed to make the measurements meaningful. In practice, many traders would get the most value from it after the first wave of genuine activity, for example somewhere between roughly 07:00 and 09:25 New York time.

The script can technically continue to display the latest calculated pre-market values after the market opens, but it is designed around pre-market accumulation logic. Its truest use case is before the open, not halfway through the regular session.

The best intervals are usually 5-minute or 15-minute. The code allows any intraday timeframe up to 30 minutes, but shorter intraday intervals generally make the pre-market path and range statistics more informative.

Operational sequence
First gather candidate movers. Then run this script on that candidate universe. Finally filter for InPlay = 1 and Avoid = 0, sort by Tradability descending, and take your top names. For the avoid list, filter Avoid = 1 and sort by AvoidScore descending.

What to Expect as Output

The script outputs numeric columns rather than a finished narrative report. Those columns are meant to be used inside a TradingView scanning workflow. The most important outputs are the following.

Output	Format	How to use it
InPlay	1 or 0	Whether the symbol passed the proxy gate.
Tradability	0-100 style score	Higher is better. Intended for ranking your tradable names.
Avoid	1 or 0	Whether the symbol belongs in the Do Not Trade bucket.

AvoidScore	0-100 style penalty score	Higher means more reasons to avoid the name.
PMDollarVol_M	Pre-market dollar volume in millions	Quick read on pre-market liquidity.
PMVolPctADV	Pre-market volume as a percent of ADV	Shows how heavy pre-market activity is relative to normal volume.
GapPct	Gap from previous close	Quick read on how large the move is.
Liquidity	0-100 style subscore	Internal component of tradability.
CatalystProxy	0-100 style subscore	Price-action stand-in for catalyst clarity.
CleanLevels	0-100 style subscore	How orderly the setup looks relative to nearby daily anchors.
Risk_* columns	0 or 1	Specific explanations for why a name is risky or belongs in avoid.
PM_Live	1 or 0	Whether the current bar is still inside the configured pre-market session.

To produce your final watchlists, use the outputs this way:

- Tradable list: InPlay = 1, Avoid = 0, then sort Tradability from highest to lowest and take the top eight.
- Avoid list: Avoid = 1, then sort AvoidScore from highest to lowest and take the top five.
- Explanation: Use the Risk_* columns to explain why a stock landed in the avoid bucket.

That means the script's final output is best thought of as a ranked decision table. It does not print paragraphs, thesis statements, or headline-based catalyst notes for each ticker. Its output is structured data.

How Changing Inputs Will Change Outcomes

Threshold tuning matters. Small changes can shift the candidate pool in meaningful ways.

Change	Likely effect
Raise minPmPctAdv from 10% to 15% or 20%	You will get fewer names, but they will tend to have more convincing early participation.
Raise minPmDollarVol from $1M to $2M+	The list becomes more liquid and more institutionally tradable, but you may lose valid small-cap momentum names.
Raise avoidPmDollarVol from $300k to $500k or $1M	The avoid bucket grows quickly because weak pre-market liquidity is heavily penalized.

Raise avoidPrice from $2 to $5	More low-priced names get filtered out. This generally reduces spread risk but may remove many active small caps.
Shorten advLen from 20 to 10	ADV becomes more reactive to recent changes in trading activity. Recent hot names may pass more easily.
Lengthen advLen from 20 to 50	ADV becomes smoother and harder to distort with a few recent outlier days. The proxy becomes more conservative.

Practical tuning advice

If your watchlist is too large and noisy, tighten the proxy thresholds first. If your watchlist contains too many cheap, thin, or spread-prone names, raise the avoid thresholds next. Those two levers usually produce the biggest quality improvement fastest.

Limitations and Important Caveats

- The script does not use a true third-party RVOL field. It uses your prompt's proxy branch instead.
- The script does not read news headlines, filings, or social feeds. Its 'catalyst clarity' score is a market-structure proxy, not a real catalyst detector.
- The script does not automatically discover the whole market. It only scores the symbols you feed into the scan.
- The script uses proxies for spread risk rather than a live bid/ask spread measurement.
- The script is specifically structured around pre-market logic. It is not a complete regular-hours trading model.
- Several meaningful thresholds are hard-coded in the source. If you want different scoring behavior, some edits require code changes rather than only input changes.

These caveats are not flaws so much as design boundaries. The script is strongest when used for exactly what it was built to do: narrow down an already-identified pool of pre-market movers into a more tradable shortlist.

Suggested Daily Use Checklist

1. Assemble a candidate list of pre-market movers.
2. Run the script on a 5m or 15m interval during pre-market.
3. Filter to InPlay = 1 and Avoid = 0.
4. Sort by Tradability and review the top eight names.
5. Filter to Avoid = 1 and sort by AvoidScore to identify your five avoid names.
6. Review the Risk_* flags to understand why a name is being rejected.

7. Use your own chart review and catalyst verification before trading. This script is a filter, not a substitute for judgment.

Final Assessment

This Pine Script is a practical pre-market triage engine. It is good at answering, in a systematic way, whether a stock looks active enough to matter, whether it appears liquid and orderly enough to trade, and whether it should instead be isolated as a low-quality setup.

Its greatest strengths are structure and consistency. Its biggest limitations are that it relies on price-and-volume proxies rather than direct news or spread data, and that it depends on you supplying the candidate universe. Used in that context, it should materially speed up the process of building a pre-market watchlist.

Most important takeaway

Treat the script as a disciplined ranking framework for pre-market candidates. The top eight names and bottom five avoids come from filtering and sorting its outputs, not from a built-in market-wide ticker picker.